



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical
Sciences



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Morphology and Word Formation in Natural Languages

Course Code & Name: DSA0305-Natural Language Processing

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Introduction



- Morphology is the branch of linguistics that studies the internal structure and formation of words.
- Every word can be broken down into morphemes, the smallest units of language that carry meaning.
- Morphemes combine systematically to build new words and to mark grammatical categories.
- Understanding word formation explains how languages expand their vocabulary over time.
- Morphological analysis also underlies key tools in computational linguistics, such as spell-checkers and machine translation.



Objectives



- To define morphology and explain its role within the broader study of linguistics.
- To distinguish between free and bound morphemes, and between roots and affixes.
- To differentiate inflectional morphology from derivational morphology.
- To examine the major word formation processes used across languages.
- To highlight how morphological structure varies across languages and supports NLP applications.



Morphemes: Free & Bound



- A morpheme is the smallest unit of language that carries meaning or grammatical function.
- Free morphemes stand alone as complete words, such as book, run, and happy.
- Bound morphemes cannot stand alone and must attach to another morpheme, such as -s, un-, and -ed.
- The root of a word carries its core meaning, for example friend in friendship.
- Affixes — prefixes, suffixes, infixes, and circumfixes — attach to roots to modify meaning or grammar.
- Example: unbelievable breaks down into un- (prefix) + believ (root) + -able (suffix).

Inflectional vs Derivational Morphology

Inflection

Derivation

Effect

Same word, different grammatical form

Creates a new word, often a new part of speech

Word class

Never changes (verb stays a verb)

Frequently changes (happy → happiness)

Examples

walk → walks, walked, walking

happy → happiness, unhappy, happily

Productivity

Highly regular and predictable

Less predictable; meaning can shift

Word Formation Processes

- Affixation – attaching a prefix or suffix to a base: happy → unhappy.
- Compounding – combining two free morphemes: note + book → notebook.
- Reduplication – repeating part or all of a word: wishy-washy.
- Conversion – shifting word class with no change in form: to google.
- Blending – merging parts of two words: smoke + fog → smog.
- Clipping – shortening a longer word: advertisement → ad.
- Initialisms – forming words from initial letters: NASA, laser.
- Borrowing and coinage – adopting or inventing new words: bungalow, Google.



Conclusion



- Morphology reveals the systematic rules by which words are built, broken down, and related to one another.
- The distinction between inflection and derivation, along with processes like affixation and compounding, explains how languages grow their vocabulary.
- Cross-linguistic typology shows that languages differ widely in how much grammatical information they pack into a single word.
- These principles are not just theoretical — they form the foundation of modern NLP systems and language technology.

Thank You